Report 2: UML Class Diagram Report

Hospital Management System UML Class Diagram Assignment | ULK | October 2026

1. Classes, Attributes and Methods

Hospital

Attributes: hospitalId: int; name: String; address: String; phone: String Methods: addDepartment(); registerPatient(); generateReport()

Department

Attributes: deptId: int; name: String; location: String Methods: addDoctor(); addRoom()
Doctor
Attributes: doctorId: int; fullName: String; specialization: String; phone: String; email: String Methods: examinePatient(); prescribeMedication(); updateRecord()

Patient

Attributes: patientId: int; fullName: String; dateOfBirth: Date; gender: String; phone: String; email: String; address: String
Methods: register(); bookAppointment(); viewMedicalRecord()

Appointment

Attributes: appointmentId: int; appointmentDate: Date; appointmentTime: Time; status: String
Methods: schedule(); cancel(); complete()

MedicalRecord

Attributes: recordId: int; diagnosis: String; treatment: String; createdDate: Date Methods: updateDiagnosis(); addTreatment(); viewHistory()

Prescription

Attributes: prescriptionId: int; medication: String; dosage: String; instructions: String; issueDate: Date
Methods: addMedication(); validate()

Room

Attributes: roomId: int; roomType: String; availabilityStatus: bool; dailyRate: double Methods: assignPatient(); releaseRoom(); checkAvailability()

Invoice

Attributes: invoiceId: int; invoiceDate: Date; totalAmount: double; paymentStatus: String Methods: calculateTotal(); processPayment(); printInvoice()

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2. Relationships and Multiplicities

Hospital Department [Composition 1 1..*]

A Hospital is made of Departments; if the hospital closes, its departments disappear too.

Hospital Patient [Aggregation 1 *]

The Hospital registers many Patients; patients can exist independently of the hospital record.

Department Doctor [Composition 1 1..*]

Each Department employs at least one Doctor; doctors belong to exactly one Department.

Department Room [Composition 1 *]

A Department has many Rooms; rooms exist only within a department.

Patient MedicalRecord [Composition 1 1]

Every Patient owns exactly one MedicalRecord; the record cannot exist without its patient.

Patient Appointment [Association 1 *]

A Patient books many Appointments; each Appointment belongs to one Patient.

Appointment Doctor [Association * 1]

Each Appointment is conducted by exactly one Doctor; a Doctor conducts many Appointments.

Doctor Prescription [Association 1 *]

A Doctor writes many Prescriptions; each Prescription is written by one Doctor.

Appointment Prescription [Association 1 0..1]

An Appointment results in zero or one Prescription; a Prescription comes from one Appointment.

Patient Invoice [Association 1 *]

A Patient receives many Invoices over time; each Invoice belongs to one Patient.

Doctor MedicalRecord [Association 1 *]

A Doctor updates many MedicalRecords; a record may be updated by many doctors over time.

3. Legend of Relationship Types

Composition (filled diamond): strong whole-part relationship the part cannot exist without the whole (e.g., Department cannot exist without its Hospital).
Aggregation (open diamond): whole-part relationship where the part can exist independently (e.g., a Patient exists outside the hospital registry).
Association (plain line): a general structural connection between two classes, annotated with a verb and multiplicities.